Shem Yu

Senior AI Engineer | AI Agents, Multimodal Systems & GenAI Platforms

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PROFESSIONAL SUMMARY

Senior AI engineer with 6+ years in applied AI. Production agents, RAG, MLOps, and shared GenAI infrastructure — serving, APIs, deploy — concentrated in the last four years, across financial services and consumer products. Hands-on on architecture and production; previously led an MLE team. Usual levers: evaluation, latency, cost, and delivery speed.

PROFESSIONAL EXPERIENCE

Senior AI Engineer | **Cookpad** Feb 2026 - Present

Develops a multimodal cooking coach that uses video and learner voice to identify where a cook is stuck and guide the next step.

- Built and iterated a staged video-understanding pipeline spanning observation, ingredient-state reasoning, and cooking-issue detection; improved issue recall against expert-annotated ground truth from 50% to 95%.
- Established capability-based evaluations and automated scoring for observation accuracy, issue coverage, factuality, coherence, and turn-level coaching quality.

Machine Learning Engineer / Team Lead | **Cathay Financial Holdings** Sep 2022 - Jan 2026

Led 4 full-time engineers (7 including contractors) across production AI deployment and internal agent development.

- Led enterprise GenAI services (AI Gateway, guardrails, MLflow) from requirements to deployment; redesigned guardrails with regex and parallel classifiers, reducing guardrail latency by roughly 60% from memory.
- Designed and delivered internal research automation that reduced a two-hour analysis workflow to 15 minutes.
- Mapped a regulatory-comparison workflow with legal and compliance stakeholders, then productionized the PoC on Databricks; improved the agent's F1 from 0.67 to 0.89. Adopted by 2 of 5 subsidiaries.

Senior AI Platform Engineer | **Wisers Information Limited** Nov 2020 - May 2022

- Standardized ML delivery with FastAPI, CI/CD, and Kubernetes templates, cutting deployment time from 2 weeks to 3 days; adopted by 30+ projects.
- Built a shared NLP Python library with 70% internal adoption and 90%+ code coverage, improving developer productivity and reliability.

Data Scientist | **TripSaaS** Dec 2019 - Nov 2020

- Analyzed customer-order data with FP-Growth and K-means clustering to support marketing segmentation.

TECHNICAL SKILLS

AI & Agent Systems	Agent System Design, Multimodal AI, Retrieval-Augmented Generation (RAG), Agent Evaluation, Embedding Databases
Engineering	Python, FastAPI, Google ADK, LangGraph, SQL, PostgreSQL, Redis
Platforms & MLOps	AWS, Databricks, Kubernetes, Docker, MLflow, CI/CD (Azure DevOps), AI Gateway, Guardrails

EDUCATION & CREDENTIALS

M.S. in Computer Science & Information Engineering, Ming Chuan University Jun 2017 - Jul 2019

AWS Certified Machine Learning - Specialty, Amazon Web Services Oct 2024

Recognition: Departmental MVP, Q1 2025; CFH Cloud Creative Award 2024, 1st Place